

Neural Particle Method: Experimental Record

Synthetic-data setup, machinery, and the benchmark suite

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Abstract

This note records, in one place, the synthetic markets and dynamics used throughout the project, the simulation, estimation and scoring machinery, and the specification of the clean benchmark suite agreed on 8 September 2026 (Section 4), run on 9 September 2026; its tables are generated from the MLflow store. The suite follows the discovery, in the diagnostics of 8 September, that all one-pass schemes were fitting the conditional expectation of the raw Euler variance rather than of its positive part, which the dynamics actually use (Section 3.4); the diagnostic runs themselves live in the MLflow experiments `mechanism`, `bump_floor` and `bump_floor_nn`.

Contents

1	Conventions	1
2	Synthetic data	2
2.1	Target surfaces: SSVI power law	2
2.2	Variance dynamics	2
2.3	Quotes, bump, and scoring	2
3	Machinery	3
3.1	Simulation	3
3.2	One-pass schemes and their estimators	3
3.3	Implicit loop	4
3.4	The variance-target discrepancy	4
4	Benchmark suite	4
4.1	Common setup	4
4.2	Cold suite	5
4.3	What a lag is	5
4.4	Lagged suite	6
4.5	Per-scenario numbers	8
4.6	Reproducibility	9

1 Conventions

Explicit denotes the one-pass, per-slice calibration with a per-slice neural network. Nadaraya–Watson (NW) is the classical particle method: the same one-pass structure with kernel regression. Other per-slice estimators are named by their estimator (RKHS, spline, ridge). *Implicit* denotes the damped outer iteration with a single network in (t, x) . All errors are implied-volatility root-mean-square errors in volatility basis points against the target surface, unless stated otherwise. “Pooled” means over all quoted strikes and maturities together; “wings” means the quoted strikes with $|k| > 0.25$; per-maturity columns restrict to one maturity.

2 Synthetic data

2.1 Target surfaces: SSVI power law

Each scenario's market is an SSVI implied-variance surface in the power-law parameterisation. With log-moneyness $k = \log(K/S_0)$, $S_0 = 1$, and parameters $(\sigma_0, \eta, \gamma, \rho_s)$,

$$\theta_T = \sigma_0^2 T, \quad \varphi(\theta_T) = \eta \theta_T^{-\gamma}, \quad w(k, T) = \frac{1}{2} \theta_T \left(1 + \rho_s \varphi k + \sqrt{(\varphi k + \rho_s)^2 + 1 - \rho_s^2} \right), \quad (1)$$

where w is total implied variance and the implied volatility is $\sqrt{w(k, T)/T}$. Parameters are drawn independently per scenario from

$$\sigma_0 \sim U(0.12, 0.35), \quad \eta \sim U(0.5, 2.0), \quad \gamma \sim U(0.3, 0.5), \quad \rho_s \sim U(-0.85, -0.10),$$

resampled until the sufficient no-arbitrage conditions $0 < \gamma \leq \frac{1}{2}$, $|\rho_s| < 1$, $\eta(1 + |\rho_s|) \leq 2$ hold. Scenario s_i uses the seed $1000 + i$ (`bench/scenarios.py`, `make_registry`).

The Dupire local volatility is analytic from Equation (1): with w_k, w_{kk}, w_T the derivatives of w ,

$$\sigma_{\text{Dup}}^2(T, k) = \frac{w_T}{1 - \frac{k}{w} w_k + \frac{1}{4} \left(-\frac{1}{4} - \frac{1}{w} + \frac{k^2}{w^2} \right) w_k^2 + \frac{1}{2} w_{kk}}, \quad (2)$$

with the denominator floored at 0.05, the local variance clipped to $[10^{-8}, 9]$, and the time argument clipped to $[t_{\min}, T]$ with $t_{\min} = 0.004$ (`market/local_vol.py`). These clipping constants are part of the calibrated behaviour and are never changed.

2.2 Variance dynamics

The calibrated model is Heston-type LSV in log-spot $X = \log S$,

$$dX_t = -\frac{1}{2} L^2(t, X_t) V_t dt + L(t, X_t) \sqrt{V_t} dW_t, \quad dV_t = \kappa(\theta - V_t) dt + \xi \sqrt{V_t} dB_t, \quad d\langle W, B \rangle_t = \rho dt. \quad (3)$$

For scenario s_i the dynamics parameters are drawn from the same seed as the surface: $\kappa \in \{1, 2, 3\}$, $\xi \in \{0.3, 0.6, 1.0\}$, $\rho \in \{-0.7, -0.3\}$, each uniform over its set, and $\theta = v_0 = \sigma_0^2$ so that the variance level matches the surface's short-end level. The Feller ratio $2\kappa\theta/\xi^2$ therefore ranges from 0.05 to 4.3 across the grid. Table 1 lists all twenty scenarios. Horizon $T = 2$ years; spot $S_0 = 1$; no rates or dividends.

Two further families exist in the registry and were not used today: six variants of s01 with (ξ, ρ) swept over the grid (`fig3_registry`), and three Heston-market scenarios whose target surface is generated by a Heston model with parameters different from the calibrated dynamics (`li_simple`, `li_complex`, `bayer`; horizon 1 year; Dupire surface from prices on a 50×121 grid).

2.3 Quotes, bump, and scoring

Quoted strikes are 13 log-strikes equally spaced in log between 0.6 and 1.6; maturities are 0.25, 0.5, 1 and 2 years. The wings are the strikes with $|k| > 0.25$ (six of thirteen). The warm-start experiments bump the surface to $\sigma'_0 = \sigma_0 + 0.01$, $\eta' = 0.95\eta$, $\rho'_s = \min(\rho_s + 0.03, -0.05)$, a level rise of one volatility point with curvature and skew nudged inside the no-arbitrage region.

Scoring reprices vanilla calls under the calibrated leverage field on fresh paths: 500 000 particles, 200 Euler steps over $[0, 2]$, seed 10 000 plus a strategy-specific offset, call price as the sample mean of the payoff, Black–Scholes inversion at the grid-snapped maturity, and the exact SSVI implied volatility at the same snapped time as the target (`pricing/reprice.py`, `pricing/metrics.py`). The bump experiments use 300 000 repricing paths. Errors are reported as RMSE in volatility bp, pooled, wings-only, and per maturity.

sid	SSVI surface				dynamics			Feller
	σ_0	η	γ	ρ_s	κ	ξ	ρ	
s01	0.261	0.52	0.34	-0.21	1	0.3	-0.7	1.51
s02	0.208	1.04	0.45	-0.56	3	0.6	-0.7	0.72
s03	0.163	0.89	0.40	-0.35	3	1.0	-0.7	0.16
s04	0.120	0.70	0.33	-0.56	2	0.6	-0.3	0.16
s05	0.285	1.01	0.49	-0.72	2	1.0	-0.3	0.32
s06	0.304	1.00	0.38	-0.58	3	1.0	-0.3	0.55
s07	0.202	0.56	0.46	-0.49	3	1.0	-0.3	0.25
s08	0.158	1.72	0.34	-0.11	1	1.0	-0.7	0.05
s09	0.133	0.99	0.31	-0.64	3	1.0	-0.3	0.11
s10	0.142	1.30	0.48	-0.11	3	1.0	-0.7	0.12
s11	0.144	1.10	0.38	-0.11	2	1.0	-0.7	0.08
s12	0.191	1.01	0.33	-0.22	3	1.0	-0.7	0.22
s13	0.303	0.62	0.43	-0.51	2	1.0	-0.3	0.37
s14	0.220	1.38	0.48	-0.11	1	0.6	-0.3	0.27
s15	0.127	0.72	0.37	-0.33	3	0.3	-0.3	1.08
s16	0.172	1.16	0.37	-0.42	1	0.6	-0.7	0.17
s17	0.136	1.13	0.42	-0.50	1	0.3	-0.3	0.41
s18	0.257	1.18	0.47	-0.24	3	0.6	-0.3	1.10
s19	0.170	1.09	0.39	-0.23	2	0.3	-0.7	1.28
s20	0.253	0.93	0.32	-0.32	3	0.3	-0.7	4.27

Table 1: The twenty synthetic scenarios: SSVI surface parameters, variance dynamics, and the Feller ratio $2\kappa\theta/\xi^2$.

3 Machinery

3.1 Simulation

Calibration and repricing share one stepper. With $\Delta = T/K$, $v^+ = \max(v, 0)$, $z_1 = \rho z_b + \sqrt{1 - \rho^2} z_p$ and independent standard normals z_b, z_p ,

$$x_{k+1} = x_k - \frac{1}{2} L_k(x_k)^2 v_k^+ \Delta + L_k(x_k) \sqrt{v_k^+} \sqrt{\Delta} z_1, \quad v_{k+1} = v_k + \kappa(\theta - v_k^+) \Delta + \xi \sqrt{v_k^+} \sqrt{\Delta} z_b. \quad (4)$$

This is full truncation: the raw v may be negative and is carried, but only v^+ enters any coefficient. Calibration uses $K = 50$ steps unless stated; repricing always uses 200. The leverage is stored per slice on a grid in log-spot and evaluated by linear interpolation, constant beyond the grid ends; the slice in force at time t is the last grid time $\leq t$. The explicit and NW passes use a per-slice grid at the 0.1% to 99.9% quantiles of the current cloud (101 points); the implicit loop uses a fixed grid of 81 points from log 0.4 to log 2.2. The leverage is capped at $L_{\max} = 4$ and the fitted conditional expectation is floored at 10^{-4} .

3.2 One-pass schemes and their estimators

At slice k , fit $f_k(x) \approx \mathbb{E}[V_{t_k} | X_{t_k} = x]$ on a subsample of 30 000 particles from the current cloud, set $L_k = \text{clip}(\sigma_{\text{Dup}}(t_k, \cdot) / \sqrt{\max(f_k, 10^{-4})}, 0, L_{\max})$, and advance the cloud. Slice 0 uses $f_0 \equiv v_0$. Estimators:

- **NW:** Gaussian Nadaraya–Watson with Silverman bandwidth $h = 1.06 \hat{\sigma}_x n^{-1/5}$ on the subsample.
- **Explicit (per-slice network):** a $1 \rightarrow 64 \rightarrow 64 \rightarrow 1$ SiLU network with softplus output scaled by 0.04, input scaled by 0.3, trained by Adam at learning rate 10^{-2} on the unweighted squared loss; 400 steps on the first fitted slice and 120 steps on each later slice, warm-started from the previous slice’s weights.

- **RKHS**: kernel ridge with a Gaussian kernel of variance 0.1 in log-spot, 100 centres at the $j \cdot 100/101$ percentiles of the cloud, regularisation $\lambda = 10^{-9}$ (Bayer et al.’s published choices).
- **Ridge head on a frozen body**: the implicit network’s body (below) as a 65-dimensional feature map, with a per-slice ridge readout, $\lambda = 10^{-3}$ per sample, shrunk toward the previous slice’s coefficients.

3.3 Implicit loop

A single network $f_\theta(t, x)$, $2 \rightarrow 64 \rightarrow 64 \rightarrow 1$ SiLU with softplus output scaled by 0.04, inputs $(t/T, x/0.3)$. Starting from the explicit output as L^0 , each outer iteration simulates a full cloud under L^n with fresh random numbers, draws a pooled subsample of 60 000 points across all slices, trains θ for 300 Adam steps at learning rate 10^{-2} with the network and optimiser state carried over from the previous iteration, reads $\Phi(L^n)_k = \text{clip}(\sigma_{\text{Dup}}(t_k)/\sqrt{\max(f_\theta(t_k, \cdot), 10^{-4})}, 0, L_{\text{max}})$ on the fixed grid, and sets $L^{n+1} = (1 - \alpha)L^n + \alpha\Phi(L^n)$ with $\alpha = \frac{1}{2}$. The paper uses six iterations. The pooled data are truncated, v^+ , before training.

3.4 The variance-target discrepancy

The one-pass schemes hand the estimator the raw Euler v ; the implicit loop hands it v^+ . Since only v^+ drives the dynamics Equation (4), the quantity the leverage must divide by is $\mathbb{E}[V^+ | X]$. Under Feller violation a sizeable fraction of particles has $v < 0$ after a coarse step (on s11, $v_0 = 0.02$, $\xi = 1$, $\Delta = 0.04$: the one-step standard deviation of v is 0.028), so $\mathbb{E}[v | X] < \mathbb{E}[v^+ | X]$ by the mean negative excursion, the denominator is too small, and the leverage too large. The effect is independent of the particle count and shrinks with the step size. “Floor on” below means the estimator is given v^+ (`ExplicitConfig.fit_v_floor`); the default remains off so that the golden replays of the paper’s runs are unchanged.

4 Benchmark suite

This section specifies the clean benchmark agreed on 8 September 2026 and run on 9 September 2026 with the variance floor of Section 3.4 as the package default. Every cell of the tables below is one MLflow run, keyed by method, scenario, seed and (for the lagged suite) lag type and offline particle count, in experiments `suite_cold`, `suite_offline`, `suite_lagged` and `pde_reference`. Every trained object is stored as a run artifact (network weights as a `torch` state dictionary, per-slice estimators and leverage fields as pickles and JSON) and can be reloaded by run id. The tables are generated from the store by `nparticle suite tables`.

4.1 Common setup

- **Scenarios**. The 20 SSVI scenarios of Table 1 and the three Heston-market scenarios used in the literature comparisons (`li_simple`, `li_complex`, `bayer`; a Heston market surface, a different Heston variance process as the calibrated model, one-year maturity, log-strikes on $[-0.9, 0.9]$). The Heston block is reported separately because it scores a single one-year maturity (the same 13 quoted strikes).
- **Time grid**. 200 Euler steps for calibration and 200 for repricing, $\Delta = 0.01$ on $[0, 2]$, so that every method sits at its step-count plateau (the step-count diagnostics of 8 September 2026, MLflow experiment `mechanism`) and the comparison is about the estimator alone.
- **Particles**. 100 000 particles for every cold row and for every online stage. Offline bodies at 200 000 and 500 000 particles.
- **Seeds**. Two seeds (0, 1) for every cold row and every online stage; one seed for each offline body. The reprice seed is 10 000 plus the calibration seed.

- **Floor.** All one-pass schemes fit $\mathbb{E}[V^+ | X]$; `fit_v_floor` becomes the default in this work and the tiny golden replays are regenerated in the same commit.
- **Implicit configuration.** Warm start from the explicit network field, 30 damped iterations at $\alpha = \frac{1}{2}$ with fresh noise per iteration, persistent network and optimiser (the best practical setting found in the convergence diagnostics of 8 September 2026, experiment `mechanism`).
- **Scoring.** As in Section 2.3: 500 000 repricing paths, Black–Scholes inversion at the snapped maturity, exact SSVI (or semi-analytic Heston) implied volatility as the target. Metrics per run: pooled and wings MAE and RMSE in volatility bp, both per maturity, the count of failed inversions, and the leverage RMSE against the PDE reference of the scenario.
- **Latency.** Wall-clock seconds of the calibration only (repricing excluded). For the lagged suite, only the online stage is timed.
- **PDE reference.** The Fokker–Planck reference leverage (`reference/pde.py`) is solved per scenario at 200 steps and enters twice: repriced under the same Monte Carlo as every other row, which is the attainable floor of the scoring under this repricer (Gyöngy), and as the reference for the leverage-error column. Its repricing is repeated under a second reprice seed to measure the Monte Carlo noise floor of the score itself.

4.2 Cold suite

One calibration from scratch per method, scenario and seed at 100 000 particles. Methods, with their estimator of $\mathbb{E}[V^+ | X]$ per slice:

1. **NW:** Gaussian Nadaraya–Watson, Silverman bandwidth (Section 3.2).
2. **Explicit NN:** per-slice network, warm-started across slices (Section 3.2).
3. **Implicit NN:** damped global-network loop (Section 3.3, configuration above).
4. **RKHS:** Gaussian kernel ridge with 100 quantile centres, Bayer et al.’s published knobs (Section 3.2).
5. **Spline:** penalised cubic B-spline, 25 quantile knots, $\lambda = 1$.
6. **Literature block.** GHL kernel: Nadaraya–Watson in spot with the quartic kernel and the Guyon–Henry–Labordère rule-of-thumb bandwidth $h(t) = 1.5 S_0 \sigma_{LV}(S_0, t) \sqrt{\max(t, 0.25)} N^{-1/5}$. Bins: piecewise-constant regression on 20 equal-frequency buckets (van der Stoep et al.). Muguruza: the kernel-free conditional Monte Carlo of Corollary 4.1, each particle contributing its one-step Gaussian conditional law of log-spot given the variance path. PU-RBF: Hakala’s partition-of-unity radial basis functions, 40 centres, 5-nearest-neighbour widths, regulariser 0.2.
7. **PDE:** the reference leverage repriced by Monte Carlo (attainable floor).

4.3 What a lag is

An overnight calibration is done on surface S_0 ; the intraday market is S_1 , and the question is how much of the overnight work survives. Because the synthetic surfaces are parametric, a lag is a move in parameter space and never a bump of quoted points, so arbitrage cannot be introduced by accident. Two lags are run for every cell.

Surface lag. For SSVI, the existing bump: $\sigma'_0 = \sigma_0 + 0.01$, $\eta' = 0.95\eta$, $\rho'_s = \min(\rho_s + 0.03, -0.05)$, shrunk by quarters until the Gatheral–Jacquier sufficient conditions ($\eta(1 + |\rho_s|) \leq 2$, $0 < \gamma \leq \frac{1}{2}$) hold; calendar no-arbitrage is automatic since $\theta_T = \sigma_0^2 T$ is increasing. For a Heston market, the matching move in vol terms: $\sqrt{v_0}$ and $\sqrt{\theta}$ up one volatility point, $\rho \rightarrow \min(\rho + 0.03, -0.05)$, $\xi \rightarrow 0.95\xi$; the surface is model-generated and arbitrage-free for any valid parameters.

Surface plus spot lag. The surface lag together with a sticky-strike spot move of +2%, $S_1 = S_0 e^\delta$, $\delta = \log 1.02$. Under sticky-strike the implied volatility at each absolute strike is unchanged, so in log-

Table 2: Cold suite, SSVI scenarios: mean and median over the 20 scenarios and two seeds. The PDE row is the attainable floor of the scoring; its seed-to-seed spread (median over scenarios of the absolute difference in pooled MAE between the two reprice seeds) is 6.6 bp, the Monte Carlo noise floor of the score.

method	pooled MAE (bp)		pooled RMSE (bp)		wings MAE (bp)		lev. RMSE	latency (s)
	mean	median	mean	median	mean	median	median	median
NW	46	35	87	67	69	50	0.101	2.4
Explicit NN	55	37	109	67	84	48	0.091	342.0
Implicit NN	52	43	103	82	80	61	0.397	566.0
RKHS	86	76	136	110	117	95	0.289	3.6
Spline	46	34	88	61	71	44	0.100	0.9
GHL kernel	46	34	89	60	71	42	0.134	2.1
Bins	63	55	118	94	100	77	0.209	0.4
Muguruza	155	143	189	173	147	138	0.170	3.1
PU-RBF	52	34	101	61	82	45	0.114	0.8
PDE (attainable floor)	42	27	85	59	62	35	0.000	0.0

Table 3: Cold suite, Heston-market scenarios (one-year maturity, same 13 quotes): mean and median over the three scenarios and two seeds..

method	pooled MAE (bp)		pooled RMSE (bp)		wings MAE (bp)		lev. RMSE	latency (s)
	mean	median	mean	median	mean	median	median	median
NW	192	67	218	71	123	68	0.164	2.4
Explicit NN	187	56	212	61	117	58	0.154	346.2
Implicit NN	181	59	209	66	108	57	1.128	572.2
RKHS	221	77	249	87	158	95	0.270	3.7
Spline	192	60	217	64	123	62	0.180	0.8
GHL kernel	191	58	216	63	122	59	0.163	2.5
Bins	194	70	223	78	126	79	0.357	0.4
Muguruza	383	466	411	500	293	403	0.766	2.5
PU-RBF	193	58	218	63	122	59	0.160	0.8
PDE (attainable floor)	189	40	211	44	123	43	0.000	0.0

moneyiness relative to the new spot the surface is translated: $\sigma'_{IV}(k, T) = \sigma_{IV}(k + \delta, T)$. A translation in k preserves Durrleman’s butterfly condition (it involves only w and its k -derivatives at a point) and calendar monotonicity, so the shifted surface is arbitrage-free. Call prices at fixed strike change with the spot, so the Dupire local volatility is that of the translated surface, $\tilde{w}(k) = w(k + \delta)$ with k measured from the new forward; the particles also start at the new spot. The stored denominators $f(t, x) = \mathbb{E}[V_t \mid X_t = x]$ were fitted for a cloud started at S_0 , and a conditional expectation given spot depends on the starting point: with $\rho < 0$ a particle sitting 2% above the old start had its variance drift down, so the stale f at the new spot is below v_0 , the stale leverage is too high near the money, most at short maturities. Sticky-delta, by contrast, leaves the surface in moneyiness unchanged and the whole calibration is scale invariant, so a sticky-delta spot move is invisible to every method; the surface-only lag already covers it. A 2% daily move is one to one-and-a-half standard deviations at 20–30% volatility.

4.4 Lagged suite

Offline stage (experiment `suite_offline`, one run per scenario, body type and particle count): a full calibration on S_0 at 200 000 or 500 000 particles, 200 steps, seed 0: either the explicit NN pass (the per-slice networks and the per-slice denominators f_k are saved) or the implicit loop (the global network is saved). These runs are cached and reused by every online cell below.

Online stage (experiment `suite_lagged`, one run per scenario, method, offline size, lag and online seed): given the intraday surface S_1 and 100 000 fresh particles, build the intraday leverage and

reprice. Only this stage is timed. Methods:

1. **Explicit NN, stale f , fresh Dupire:** $L_k = \text{clip}(\sigma_{\text{Dup}}^{S_1}(t_k, \cdot) / \sqrt{\max(f_k, 10^{-4})}, 0, L_{\max})$ with the offline per-slice f_k . Costs nothing online beyond the repricing.
2. **Implicit NN, stale f , fresh Dupire:** the same with $f = f_\theta(t, \cdot)$ from the offline global network.
3. **Explicit NN with RKHS head:** simulate the 100 000 online particles under the stale- f leverage of item 1 with fresh Dupire, slice by slice; at each slice fit a Gaussian kernel ridge to the residual $v^+ - f_k(x)$ on the online cloud, and use $f_k + \text{residual}$ as the denominator for the next step (a causal one-pass correction).
4. **Implicit NN with RKHS head:** the same residual head on the global network's f_θ .
5. **Explicit NN with ridge head:** the per-slice network body frozen as a feature map (64 features plus a constant) with its last layer refitted by ridge on the online cloud, shrunk toward the offline readout. The body's output is $\text{softplus}(Aw) \cdot 0.04$, and the refit minimises $\sum_i (\text{softplus}(A_i w) \cdot 0.04 - v_i^+)^2 + \lambda_{\text{eff}} \|w - w_0\|^2$ on the raw variance scale by Gauss–Newton to convergence (at most 20 steps) from the offline last layer w_0 , with $\lambda_{\text{eff}} = \lambda$ times the mean diagonal of the Gauss–Newton curvature at w_0 and $\lambda = 10^{-3}$, so the penalty is a fixed fraction of the data curvature whatever the output scale and cloud size. The denominator is $\text{softplus}(Aw) \cdot 0.04$ at the converged w . When the online data agree with the offline fit the first step is zero and $w = w_0$.
6. **Implicit NN with ridge head:** the same on the global body in (t, x) .

The online reference rows are NW re-solved from scratch on S_1 at 100 000 particles and 200 steps (the best intraday strategy in the floor-corrected bump runs, experiment `bump_floor`), and the stale leverage with no action.

Table 4: Lagged suite, SSVI scenarios: mean over the 20 scenarios and two online seeds; latency is the online stage only. One block per lag type..

method	offline N	pooled MAE	pooled RMSE	wings MAE	$T = 0.25$ MAE	online latency (s)
<i>Surface lag</i>						
Stale leverage, no action	—	97	131	102	140	0.0
NW re-solve on S_1	—	46	84	67	98	3.7
Explicit NN, stale f , fresh Dupire	200k	55	103	78	123	0.0
Explicit NN, stale f , fresh Dupire	500k	55	103	77	122	0.0
Implicit NN, stale f , fresh Dupire	200k	72	130	104	149	0.0
Implicit NN, stale f , fresh Dupire	500k	54	97	77	115	0.0
Explicit NN + RKHS head	200k	49	89	73	102	6.1
Explicit NN + RKHS head	500k	47	86	70	99	6.2
Implicit NN + RKHS head	200k	51	90	76	98	6.2
Implicit NN + RKHS head	500k	59	99	85	108	6.2
Explicit NN + ridge head	200k	50	95	74	115	9.6
Explicit NN + ridge head	500k	50	97	75	118	9.9
Implicit NN + ridge head	200k	48	91	71	109	11.3
Implicit NN + ridge head	500k	47	89	69	105	10.7
<i>Surface plus 2% sticky-strike spot lag</i>						
Stale leverage, no action	—	95	135	106	164	0.0
NW re-solve on S_1	—	46	85	68	100	3.7
Explicit NN, stale f , fresh Dupire	200k	99	142	118	202	0.0
Explicit NN, stale f , fresh Dupire	500k	99	142	117	200	0.0
Implicit NN, stale f , fresh Dupire	200k	104	157	131	208	0.0
Implicit NN, stale f , fresh Dupire	500k	91	128	108	169	0.0
Explicit NN + RKHS head	200k	51	94	76	111	6.1
Explicit NN + RKHS head	500k	54	96	80	111	6.2
Implicit NN + RKHS head	200k	65	111	94	125	6.2
Implicit NN + RKHS head	500k	53	96	78	110	6.1
Explicit NN + ridge head	200k	48	92	72	110	9.8
Explicit NN + ridge head	500k	48	92	71	112	10.2
Implicit NN + ridge head	200k	48	90	72	102	11.5
Implicit NN + ridge head	500k	49	91	71	108	10.9

Table 5: Lagged suite, Heston-market scenarios..

method	offline N	pooled MAE	pooled RMSE	wings MAE	online latency (s)
<i>Surface lag</i>					
Stale leverage, no action	–	291	317	203	0.0
NW re-solve on S_1	–	232	257	159	3.7
Explicit NN, stale f , fresh Dupire	200k	244	267	167	0.0
Explicit NN, stale f , fresh Dupire	500k	247	270	169	0.0
Implicit NN, stale f , fresh Dupire	200k	241	266	164	0.0
Implicit NN, stale f , fresh Dupire	500k	228	254	151	0.0
Explicit NN + RKHS head	200k	233	257	160	6.1
Explicit NN + RKHS head	500k	233	257	160	6.1
Implicit NN + RKHS head	200k	234	259	161	6.1
Implicit NN + RKHS head	500k	231	255	158	6.0
Explicit NN + ridge head	200k	231	256	159	9.4
Explicit NN + ridge head	500k	233	257	160	9.6
Implicit NN + ridge head	200k	227	252	153	11.4
Implicit NN + ridge head	500k	226	252	153	9.5
<i>Surface plus 2% sticky-strike spot lag</i>					
Stale leverage, no action	–	274	306	188	0.0
NW re-solve on S_1	–	224	248	152	3.6
Explicit NN, stale f , fresh Dupire	200k	221	248	148	0.0
Explicit NN, stale f , fresh Dupire	500k	225	252	151	0.0
Implicit NN, stale f , fresh Dupire	200k	214	240	140	0.0
Implicit NN, stale f , fresh Dupire	500k	198	225	126	0.0
Explicit NN + RKHS head	200k	221	243	150	6.0
Explicit NN + RKHS head	500k	221	243	150	6.1
Implicit NN + RKHS head	200k	222	246	152	6.1
Implicit NN + RKHS head	500k	218	241	147	6.1
Explicit NN + ridge head	200k	220	244	149	9.4
Explicit NN + ridge head	500k	220	243	148	9.8
Implicit NN + ridge head	200k	216	240	143	11.5
Implicit NN + ridge head	500k	216	240	143	9.7

4.5 Per-scenario numbers

Cold suite, SSVI scenarios, pooled MAE (bp)

sid	NW	Explicit NN	Implicit NN	RKHS	Spline	GHL kernel	Bins	Muguruza	PU-RBF
s01	16	14	16	14	14	14	20	13	15
s02	14	14	22	68	13	13	19	31	27
s03	61	102	65	169	64	63	94	241	100
s04	105	101	121	108	109	104	137	190	103
s05	49	37	53	81	40	41	52	138	37
s06	23	21	28	44	24	23	33	85	22
s07	46	41	48	123	47	44	75	188	43
s08	78	124	79	119	79	76	109	436	98
s09	132	131	174	229	135	132	152	258	133
s10	31	44	61	141	28	30	57	342	31
s11	76	191	87	157	77	96	122	333	112
s12	52	48	61	80	53	52	79	229	67
s13	27	24	29	81	26	26	37	141	24
s14	25	30	19	71	26	27	26	146	29
s15	72	69	68	68	70	69	75	70	69
s16	40	36	38	71	39	39	65	164	61
s17	26	24	22	27	26	24	41	40	27
s18	18	21	18	23	19	19	18	31	20
s19	22	20	22	33	21	20	32	23	23
s20	11	9	14	10	10	10	18	9	9

Cold suite, Heston scenarios, pooled MAE (bp)

sid	NW	Explicit NN	Implicit NN	RKHS	Spline	GHL kernel	Bins	Muguruza	PU-RBF
bayer	500	499	476	519	510	510	502	659	515
li_complex	67	56	59	67	60	58	70	466	58
li_simple	9	5	8	77	6	5	9	24	5

Lagged suite, surface lag, SSVI scenarios, pooled MAE (bp)

sid	explicit_ridge/200000	explicit_ridge/500000	explicit_rkhs/200000	explicit_rkhs/500000	explicit_atale/200000	explicit_atale/500000	implicit_ridge/200000	implicit_ridge/500000	implicit_rkhs/200000	implicit_rkhs/500000	implicit_atale/200000	implicit_atale/500000	nn_resolve/0	stale_L/200000
s01	14	14	13	13	17	17	13	13	13	13	19	15	15	96
s02	16	15	20	16	21	20	14	15	18	17	21	17	15	74
s03	70	74	73	67	92	94	61	56	69	61	88	62	61	109
s04	100	95	106	105	98	95	94	104	119	104	188	116	104	132
s05	36	35	39	45	24	24	32	35	34	35	55	36	43	56
s06	23	23	23	26	22	21	23	23	24	23	86	38	24	83
s07	44	44	46	46	46	44	42	46	44	51	64	57	48	84
s08	96	112	82	86	112	126	102	126	93	87	109	143	89	120
s09	133	125	120	147	148	146	134	113	161	372	228	127	120	139
s10	33	34	33	30	41	42	38	47	40	37	58	51	36	64
s11	152	152	92	86	147	133	111	79	86	79	122	71	75	148
s12	50	50	50	50	53	53	56	54	79	54	65	58	54	91
s13	26	25	59	30	30	31	27	25	28	30	35	22	28	87
s14	30	30	29	30	42	41	30	30	28	29	29	55	24	88
s15	61	61	63	63	69	68	61	61	63	61	84	70	64	135
s16	37	36	37	38	46	47	35	36	40	49	51	57	44	77
s17	22	22	22	23	31	31	25	22	34	23	82	22	24	92
s18	21	21	21	21	22	22	22	21	21	21	13	21	18	85
s19	19	19	18	19	23	22	20	19	19	21	25	21	20	92
s20	9	9	9	9	14	13	9	9	10	9	18	13	10	91

Lagged suite, surface plus 2% spot lag, SSVI scenarios, pooled MAE (bp)

sid	explicit_ridge/200000	explicit_ridge/500000	explicit_rkhs/200000	explicit_rkhs/500000	explicit_atale/200000	explicit_atale/500000	implicit_ridge/200000	implicit_ridge/500000	implicit_rkhs/200000	implicit_rkhs/500000	implicit_atale/200000	implicit_atale/500000	nn_resolve/0	stale_L/200000
s01	14	14	14	14	63	62	14	13	14	13	59	59	15	51
s02	15	15	16	17	103	97	13	14	17	15	93	98	15	46
s03	70	71	67	69	189	185	59	73	81	77	133	117	61	110
s04	104	94	103	121	114	109	106	96	177	106	184	143	101	128
s05	34	34	56	34	46	46	31	33	32	34	89	30	39	111
s06	23	23	23	23	33	32	23	23	25	22	62	73	24	103
s07	44	44	47	44	59	56	44	43	43	66	51	58	47	92
s08	93	100	107	109	155	176	86	129	118	118	136	152	93	132
s09	123	110	171	233	159	155	148	102	314	196	238	139	119	122
s10	48	48	43	44	156	170	45	69	51	47	155	69	51	103
s11	113	131	96	97	265	251	103	106	109	111	164	115	79	218
s12	53	53	58	54	122	123	61	54	70	58	139	131	55	69
s13	25	25	26	26	37	36	27	26	28	28	42	38	28	91
s14	21	20	19	19	62	62	20	20	19	19	62	58	18	54
s15	65	65	63	64	84	83	64	64	66	63	80	85	65	133
s16	37	38	39	39	103	101	34	35	38	40	82	144	44	59
s17	23	23	24	24	57	58	25	23	35	25	103	65	24	106
s18	20	20	20	20	32	31	21	20	20	19	44	11	17	77
s19	19	19	20	20	86	83	21	20	20	20	93	82	21	40
s20	10	9	10	10	56	54	10	9	10	9	38	56	11	65

Lagged suite, surface lag, Heston scenarios, pooled MAE (bp)

sid	explicit_ridge/200000	explicit_ridge/500000	explicit_rkhs/200000	explicit_rkhs/500000	explicit_atale/200000	explicit_atale/500000	implicit_ridge/200000	implicit_ridge/500000	implicit_rkhs/200000	implicit_rkhs/500000	implicit_atale/200000	implicit_atale/500000	nn_resolve/0	stale_L/200000
bayer	620	625	626	625	634	642	606	607	624	622	613	609	612	628
li_complex	65	65	64	65	81	79	66	63	70	61	87	44	73	138
li_simple	9	9	9	9	17	19	9	9	9	9	23	33	11	108

Lagged suite, surface plus 2% spot lag, Heston scenarios, pooled MAE (bp)

sid	explicit_ridge/200000	explicit_ridge/500000	explicit_rkhs/200000	explicit_rkhs/500000	explicit_atale/200000	explicit_atale/500000	implicit_ridge/200000	implicit_ridge/500000	implicit_rkhs/200000	implicit_rkhs/500000	implicit_atale/200000	implicit_atale/500000	nn_resolve/0	stale_L/200000
bayer	596	595	595	595	532	545	585	586	595	593	505	500	595	553
li_complex	56	55	58	57	94	92	57	54	63	52	98	58	65	167
li_simple	9	9	10	9	38	38	8	9	10	9	37	38	11	104

4.6 Reproducibility

Each run records the git hash, the full configuration as parameters, the leverage field, the implied-volatility error grid, and the trained objects. A loader (`nparticle suite load <run_id>`) rebuilds the leverage field and, where present, the network or per-slice estimators from the artifacts alone, so any row of the tables can be re-scored or reused as an offline body without recomputation.